BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 062 · Programming Challenge and Review

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. A £5-per-ticket program uses total = tickets + 5. What is the correction?

Q2. Insert 6 into sorted [2, 5, 8]. What is the result?
A. [2, 5, 8, 6]
B. [2, 5, 6, 8]
C. [6, 2, 5, 8]
D. [8, 6, 5, 2]

Q3. “A comparison operator must always produce a text paragraph.” Correct this. Explain why.

Q4. A warning needs high temperature OR high pressure. “OR is false when both inputs are true.” Correct this. What happens when both are high?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Planning — Define inputs, processing, output and success criteria.

Selection — Use IF or ELSE to choose valid outcomes.

Iteration — Repeat actions with a bounded FOR or condition-controlled WHILE.

Testing — Check normal, boundary and invalid inputs, then refine.

Algorithm — A sequence of precise problem-solving instructions.

Trace table — A table tracking variable values during execution.

New learning

Planning: Define inputs, processing, output and success criteria. For example, consider a club booking tool.

Selection: Use IF or ELSE to choose valid outcomes. For example, consider a revision calculator.

Iteration: Repeat actions with a bounded FOR or condition-controlled WHILE. For example, consider a rewards app.

Testing: Check normal, boundary and invalid inputs, then refine. For example, consider a canteen voucher.

Worked example

A club booking tool is being developed. Explain how planning and selection are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Planning [1]; Define inputs, processing, output and success criteria. [1].

3. Selection [1]; Use IF or ELSE to choose valid outcomes. [1].

4. Give one accurate point and a linked reason for each technique.

We Do · Practise with guidance

Guided 1

A revision calculator is being programmed. Explain why planning matters. [2 marks]

Worked method: Planning [1]; Define inputs, processing, output and success criteria. [1].

Guided 2

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Use total = tickets * 5; addition does not calculate £5 for each ticket.

Q2. B — [2, 5, 6, 8]

Q3. A Boolean result: true or false.

Q4. OR is true if at least one input is true, including both. AND needs both true. NOT reverses a value.

I Do / We Do pages · Guided practice

Q1. Planning [1]; Define inputs, processing, output and success criteria. [1].

Q2. Selection [1]; Use IF or ELSE to choose valid outcomes. [1].

Q3. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1].

You Do pages · Code First

```
sensor = 29
if sensor > 28 then
    print("Too hot")
else
    print("OK")
endif
```

Expected result: Too hot

You Do pages · Independent questions

Q1. Planning [1]; Iteration [1].

Q2. Selection [1]; Use IF or ELSE to choose valid outcomes. [1]. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1].

Q3. Planning [1]; Define inputs, processing, output and success criteria. [1]. Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

For a rewards app, explain how selection is used. [2 marks]

Worked method: Selection [1]; Use IF or ELSE to choose valid outcomes. [1].

Guided 3

A canteen voucher needs iteration and testing. Explain the roles of both. [4 marks]

Worked method: Iteration [1]; Repeat actions with a bounded FOR or condition-controlled WHILE. [1]. Testing [1]; Check normal, boundary and invalid inputs, then refine. [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the missing condition to report a hot sensor.

```
sensor = 29
if ... then
    print("Too hot")
else
    print("OK")
endif
```

write

Question 1

State TWO concepts needed for this programming technique. [2 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 2

Explain how selection and iteration can help a fundraising tickets. [4 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 3

Explain the roles of planning, iteration and testing when creating a quiz score. [6 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A useful algorithm plan identifies inputs, processing, outputs and success criteria.